

OSM / Opto-PING Rung 0M: Null-Aware Model Selection and K-Significance Calibration

Overall status: PARTIAL_OR_FAILED_NULL_AWARE_SPECIFICITY

Rung 0M was run because Rung 0L exposed a crucial weakness: a model can win prediction under structured nulls even when the loop should not be believed. Rung 0M therefore requires two simultaneous gates: (1) full reciprocal prediction must win under held-out conditions, and (2) K must lock against null-calibrated shuffled, permuted, and time-reversed controls. Full-model prediction alone is not enough.

Spec SHA-256: 6e9be8fc28625ad63c851af513eb39195d321bef0865ccbe18311b0208e47659
Total trials: 120 | **Positive:** 20 | **Null:** 100 | **Null-calibration controls per trial:** 9
Boundary: Synthetic sensitivity/specificity calibration only; no biological, PR-AYC-G, human EEG, microtubular memory, biophotonic, or OSM mechanism claim.

Executive numerical result

Metric	Observed	Criterion	Status
Positive full + K lock rate	0.600	target >= 0.800	FAIL
Positive K significant rate	1.000	target >= 0.850	PASS
Positive K within 15% rate	1.000	target >= 0.800	PASS
Positive K within 10% rate	0.950	target high	PASS-like
Positive full model CV win rate	0.600	target >= 0.700	FAIL
Null false full + K lock rate	0.000	target <= 0.050	PASS
Null K significant rate	0.350	target <= 0.100	FAIL
Null full model CV win rate	0.062	low is better	PASS-like
Null median K_hat_scaled	0.064	near zero	PASS-like

Bottom line: Rung 0M improved specificity by eliminating false full+K locks in the aggregate null set, but it overcorrected: positive-control full+K sensitivity only reached 0.600. The gate is scientifically useful precisely because it did not let the model declare victory under an unbalanced sensitivity/specificity tradeoff.

Generator summary

generator	n_trials	median_K_hat_s caled	K_significant_rat e	full_cv_win_rate _mean	full_and_K_lock _rate	non_full_cv_win_r ate_mean	median_K_null _p
colored_noise_null	20	0.789	1.000	0.058	0.000	0.942	0.100
positive_control	20	1.007	1.000	0.600	0.600	0.400	0.100
generic_hidden_null	20	0.054	0.100	0.000	0.000	1.000	0.750
replay_only_null	20	0.037	0.050	0.042	0.000	0.958	0.600
sensory_only_null	20	0.073	0.350	0.000	0.000	1.000	0.200
standard_ping_null	20	0.045	0.250	0.208	0.000	0.792	0.600

Generator-specific decision table

generator	criterion	observed	threshold	pass
positive_control	full_and_K_lock_rate	0.600	>= 0.8	False
positive_control	K_significant_rate	1.000	>= 0.85	True
positive_control	full_cv_win_rate_mean	0.600	>= 0.7	False
positive_control	K_within_15_positive_rate	1.000	>= 0.8	True
standard_ping_null	false_full_and_K_lock_rate	0.000	<= 0.05	True
standard_ping_null	K_significant_rate	0.250	<= 0.1	False
standard_ping_null	full_cv_win_rate_mean	0.208	<= 0.35	True
replay_only_null	false_full_and_K_lock_rate	0.000	<= 0.05	True
replay_only_null	K_significant_rate	0.050	<= 0.1	True
replay_only_null	full_cv_win_rate_mean	0.042	<= 0.2	True
sensory_only_null	false_full_and_K_lock_rate	0.000	<= 0.05	True
sensory_only_null	K_significant_rate	0.350	<= 0.1	False
sensory_only_null	full_cv_win_rate_mean	0.000	<= 0.2	True
generic_hidden_null	false_full_and_K_lock_rate	0.000	<= 0.05	True
generic_hidden_null	K_significant_rate	0.100	<= 0.1	True
generic_hidden_null	full_cv_win_rate_mean	0.000	<= 0.25	True
colored_noise_null	false_full_and_K_lock_rate	0.000	<= 0.05	True
colored_noise_null	K_significant_rate	1.000	<= 0.1	False
colored_noise_null	full_cv_win_rate_mean	0.058	<= 0.25	True

Held-out cross-validation winner pattern

The full reciprocal model rarely won under null generators after stronger alternatives and penalties were introduced. This is the main specificity improvement. However, K significance remained too permissive in colored-noise and sensory-drive nulls, so K calibration remains the next repair target.

generator	winning_model	wins	total_folds	win_rate
colored_noise_null	colored_noise_ar	61	120	0.508
colored_noise_null	eta_only	44	120	0.367
colored_noise_null	full_recip	7	120	0.058
colored_noise_null	generic_hidden	2	120	0.017
colored_noise_null	replay_only	6	120	0.050
positive_control	colored_noise_ar	8	120	0.067
positive_control	eta_only	36	120	0.300
positive_control	full_recip	72	120	0.600
positive_control	standard_ping	1	120	0.008
positive_control	zeta_only	3	120	0.025
generic_hidden_null	colored_noise_ar	7	120	0.058
generic_hidden_null	generic_hidden	113	120	0.942
replay_only_null	eta_only	32	120	0.267
replay_only_null	full_recip	5	120	0.042
replay_only_null	generic_hidden	2	120	0.017
replay_only_null	replay_only	52	120	0.433
replay_only_null	standard_ping	4	120	0.033
replay_only_null	zeta_only	25	120	0.208
sensory_only_null	colored_noise_ar	4	120	0.033
sensory_only_null	eta_only	1	120	0.008
sensory_only_null	generic_hidden	3	120	0.025
sensory_only_null	sensory_only	107	120	0.892
sensory_only_null	standard_ping	4	120	0.033
sensory_only_null	zeta_only	1	120	0.008
standard_ping_null	eta_only	79	120	0.658
standard_ping_null	full_recip	25	120	0.208
standard_ping_null	standard_ping	13	120	0.108
standard_ping_null	zeta_only	3	120	0.025

Interpretation and next gate

What improved: The final lock now requires prediction + K together. Under the empty-sky/null generators, false full+K lock rate was 0.000, meaning the radar did not declare final target lock in empty skies.

What failed: The positive-control lock rate was only 0.600. The v0.1 model-selection penalties and alternatives made the gate conservative enough to reject nulls, but also conservative enough to miss many true synthetic reciprocal-loop cases. Null K significance was also too high under colored noise and sensory-drive-only generators.

Rung 0N requirement: Keep the simultaneous full+K rule, but recalibrate the K-null layer. Use higher-resolution or pooled null distributions, generator-specific K thresholds, and penalty calibration that preserves positive-control sensitivity while keeping false full+K locks near zero.

Boundary: This is still synthetic sensitivity/specificity calibration. It does not prove or disprove biological OSM, microtubular memory, biophotonic flickering, human EEG mechanisms, or PR-AYC-G empirical results.